Positivity of a family of Stern polynomials and a counterexample to monotonicity

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Abstract

For the standard Stern polynomials, we prove that $B_{h_n}(t) > 0$ for every real t and every integer $n \geq 0$, where $h_n = \frac{2}{3}(2^{2n}-1)(2^{2n+1}+1)+1$. This proves the assertion of Conjecture 4.4 in the arXiv version of Ulas's *Strong arithmetic property of certain Stern polynomials*. The proof derives from the defining recurrence a quadratic identity already present in that source, and completes it to a sum of squares whose terms cannot vanish simultaneously. We also compute B_{183} and show that its derivative at $-2/3$ is $-19/81$. An exact two-point certificate disproves the final monotonicity clause of Conjecture 4.3 even on $(-1, 0)$. The general unique-real-root clause of that conjecture is not settled here. A Lean 4 formalization certifies the all-parameter positivity theorem and the exact monotonicity counterexample from the defining recurrence. No claim of literature priority is made.

1 Definitions and the statements under consideration

Let $\mathbb{N} = \{0, 1, 2, \ldots\}$. The Stern polynomials $B_m \in \mathbb{Z}[t]$ are defined by

$$B_0(t) = 0, \qquad B_1(t) = 1, \qquad B_{2m}(t) = tB_m(t), \qquad B_{2m+1}(t) = B_m(t) + B_{m+1}(t). \tag{1}$$

The recurrence determines the polynomials when used for $m \geq 1$; both displayed identities remain valid for $m = 0$. This is the standard definition introduced by Klavžar, Milutinović, and Petr [1], and is the definition in the body of Ulas's paper [2, p. 2]. The odd-index formula in that paper's abstract contains a different subscript; the present note uses the body of the paper and (1).

The sequence in Conjecture 4.4 of [2, p. 16] is

$$h_n = \frac{2}{3}(2^{2n} - 1)(2^{2n+1} + 1) + 1 \qquad (n \in \mathbb{N}). \tag{2}$$

This is an integer sequence: if $K = 4^n$ and $a = (K - 1)/3$, then $K \equiv 1 \pmod 3$, $a \in \mathbb{N}$, and

$$K = 3a + 1, \qquad h_n = 2a(2K + 1) + 1 = 4Ka + 2a + 1. \tag{3}$$

In particular $h_0 = 1$, $h_1 = 19$, and $h_2 = 331$. We will prove the stronger form of the conjectured absence of real roots:

$$B_{h_n}(t) > 0 \qquad (n \in \mathbb{N},\ t \in \mathbb{R}). \tag{4}$$

For the monotonicity question, the family in [2, Theorem 2.6] is

$$s_{0,n} = 2^{2n+4} - 9 \cdot 2^{n+1} - 1 \qquad (n \geq 1). \tag{5}$$

The last clause of Conjecture 4.3, numbered "(3)" again in the source, asserts that $t \mapsto B_{s_{0,n}}(t)$ is increasing on $\mathbb{R}$. The value $n = 2$ lies in the range $n \geq 2$ used for the first clause, as well as in the positive-index range of the family. Thus it supplies a counterexample under either of these index conventions.

2 Two consequences of the defining recurrence

Lemma 1 (Concatenation identity)**.** *For integers $m, r, s \geq 0$ with $s \leq 2^r$, the following identity holds in $\mathbb{Z}[t]$:*

$$B_{m2^r+s} = B_{2^r-s}B_m + B_sB_{m+1}. \tag{6}$$

All polynomials on the right have the same variable t.

Proof. We induct on r. If $r = 0$, then $s = 0$ or $s = 1$; the assertion follows from $B_0 = 0$ and $B_1 = 1$. Suppose the result holds for r and put $K = 2^r$. For an even remainder $s = 2u$, where $0 \leq u \leq K$, the induction hypothesis and the even recurrence give

$$\begin{aligned} B_{2mK+2u} &= t(B_{K-u}B_m + B_uB_{m+1}) \\ &= B_{2K-2u}B_m + B_{2u}B_{m+1}. \end{aligned}$$

For an odd remainder $s = 2u + 1 \leq 2K$, we have $0 \leq u < K$. Both u and $u + 1$ are admissible remainders at level r. Consequently,

$$\begin{aligned} B_{2mK+2u+1} &= B_{mK+u} + B_{mK+u+1} \\ &= (B_{K-u} + B_{K-u-1})B_m + (B_u + B_{u+1})B_{m+1} \\ &= B_{2K-2u-1}B_m + B_{2u+1}B_{m+1}. \end{aligned}$$

Here $K - u - 1 \geq 0$, so the last use of the odd recurrence is legitimate. This proves the induction step, including both endpoints of the range. □

Lemma 2 (Adjacent values)**.** *If $t \in \mathbb{R} \setminus \{0\}$ and $m \in \mathbb{N}$, then $(B_m(t), B_{m+1}(t)) \neq (0, 0)$. At $t = 0$, every odd-indexed Stern polynomial has value 1.*

Proof. Write $v_m = (B_m(t), B_{m+1}(t))^{\mathsf{T}}$. The recurrence gives

$$v_{2m} = \begin{pmatrix} t & 0 \\ 1 & 1 \end{pmatrix} v_m, \qquad v_{2m+1} = \begin{pmatrix} 1 & 1 \\ 0 & t \end{pmatrix} v_m. \tag{7}$$

For $t \neq 0$, both matrices have nonzero determinant t and hence preserve the property of being a nonzero vector. Starting from $v_0 = (0, 1)^{\mathsf{T}}$, strong induction on the index proves the first assertion: for every positive index, its parent $\lfloor m/2 \rfloor$ is smaller.

At $t = 0$, the two transformations in (7) take a pair (x, y) with $x + y = 1$ to $(0, 1)$ or $(1, 0)$, respectively. The initial pair has sum 1, so the same induction shows that $B_m(0) + B_{m+1}(0) = 1$ for every m. The odd recurrence now gives $B_{2m+1}(0) = 1$. □

3 The quadratic identity and strict positivity

Proposition 3. *For $n \in \mathbb{N}$, let $a = (4^n - 1)/3$, $A = B_a(t)$ and $C = B_{a+1}(t)$. Then*

$$B_{h_n}(t) = (t^2 + t + 1)A^2 + (t + 2)AC + C^2. \tag{8}$$

Proof. Set $K = 4^n = 3a + 1$. Since $2a + 1 \leq K \leq 4K$ and $4K = 2^{2n+2}$, Lemma 1 applied to $h_n = a(4K) + (2a + 1)$ gives

$$B_{h_n} = B_{4K-2a-1}A + B_{2a+1}C. \tag{9}$$

The second coefficient is $B_{2a+1} = A + C$. To compute the first, the odd recurrence gives

$$B_{4K-2a-1} = B_{2K-a-1} + B_{2K-a}.$$

We have $K - a - 1 = 2a$ and $K - a = 2a + 1$. Applying Lemma 1 with $m = 1$, $2^r = K$, and remainders $2a$ and $2a + 1$, which both lie in $[0, K]$, yields

$$\begin{aligned} B_{2K-a-1} = B_{K+2a} &= B_{K-2a}B_1 + B_{2a}B_2 = C + t^2A, \\ B_{2K-a} = B_{K+2a+1} &= B_{K-2a-1}B_1 + B_{2a+1}B_2 = A + t(A + C). \end{aligned}$$

Substitution in (9) proves (8). The calculations also apply when $n = 0$, for which $K = 1$ and $a = 0$. □

Remark 4 (Attribution). Ulas already gives identity (8) in the proof of Theorem 3.1 [2, p. 13]. No novelty is claimed for that identity. The derivation above makes the dependence on the original index sequence and defining recurrence explicit. It does not use the subsequent displayed substitution or initial values in that source.

Theorem 5. *For every $n \in \mathbb{N}$ and every $t \in \mathbb{R}$, $B_{h_n}(t) > 0$. In particular B_{h_n} has no real root.*

Proof. Completing the square in (8) gives

$$4B_{h_n}(t) = \big(2C + (t + 2)A\big)^2 + 3t^2A^2. \tag{10}$$

Both terms on the right are nonnegative. If $t \neq 0$ and their sum were zero, then $3t^2A^2 = 0$ would imply $A = 0$. The other square would then imply $C = 0$. This contradicts Lemma 2 for the adjacent indices $a, a + 1$. Thus $B_{h_n}(t) > 0$ for $t \neq 0$.

When $t = 0$, formula (3) shows that h_n is odd, so Lemma 2 gives $B_{h_n}(0) = 1$. In particular the endpoint $n = 0$ is covered: $h_0 = 1$ and $B_{h_0}(t) = 1$ for all t. □

4 An exact failure of monotonicity

Proposition 6. *For the sequence* (5)*, $s_{0,2} = 183$ and*

$$P(t) := B_{183}(t) = t^5 + 3t^4 + 7t^3 + 9t^2 + 5t + 1. \tag{11}$$

Moreover,

$$P'(-2/3) = -19/81 < 0. \tag{12}$$

The function P is neither nondecreasing nor strictly increasing on $(-1, 0)$, and hence has neither property on $\mathbb{R}$.

Proof. First $s_{0,2} = 2^8 - 9 \cdot 2^3 - 1 = 183$. For transparency, the following intermediate values follow from (1):

$$\begin{aligned} B_5 &= 1 + 2t, & B_6 &= t + t^2, \\ B_{11} &= 1 + 3t + t^2, & B_{12} &= t^2 + t^3, \\ B_{22} &= t + 3t^2 + t^3, & B_{23} &= 1 + 3t + 2t^2 + t^3, \\ B_{45} &= 1 + 4t + 5t^2 + 2t^3, & B_{46} &= t + 3t^2 + 2t^3 + t^4, \\ B_{91} &= 1 + 5t + 8t^2 + 4t^3 + t^4, & B_{92} &= t^2 + 3t^3 + 2t^4 + t^5. \end{aligned}$$

Since $B_{183} = B_{91} + B_{92}$, these give (11). Differentiating,

$$P'(t) = 5t^4 + 12t^3 + 21t^2 + 18t + 5,$$

and exact substitution gives (12). Continuity of P' gives an interval around $-2/3$, contained in $(-1, 0)$, on which $P' < 0$. The mean value theorem therefore gives strict decrease on that interval.

There is also a certificate using only rational arithmetic:

$$-1 < -\frac{2}{3} < -\frac{3}{5} < 0, \qquad P(-2/3) = \frac{13}{243}, \qquad P(-3/5) = \frac{122}{3125},$$
$$P(-3/5) - P(-2/3) = -\frac{10979}{759375} < 0.$$

These two points directly contradict both meanings of increasing specified in the statement. □

Proposition 7. *The polynomial B_{183} has exactly one real root, namely -1, and that root is simple.*

Proof. Expansion verifies the identity

$$P(t) = (t + 1)\left([t(t + 1)]^2 + (2t + 1)^2\right).$$

The first square can vanish only at 0 or -1, whereas the second can vanish only at $-1/2$. Their sum is therefore strictly positive on $\mathbb{R}$. This proves that $P(t) = 0$ precisely when $t = -1$. The factor in parentheses has value 1 at -1, so the root is simple. □

Proposition 7 prevents a possible misinterpretation of the counterexample: the failure of monotonicity does not refute the unique-real-root clause of Conjecture 4.3. No assertion for the general parameters of that clause is proved here. Finite root counts do not establish that general assertion.

5 Formal verification and reproducibility

The accompanying project has been checked using Lean 4.19.0 and mathlib 4.19.0. It defines a real-valued function B recursively and proves the standard recurrence. The theorem `Stern.B_integer_polynomial` constructs, for every index, an integer polynomial whose real evaluation is B. The formal identities are pointwise equalities of real-valued functions; the project does not separately construct a canonical sequence of objects in $\mathbb{Z}[X]$ and prove coefficient equalities between those objects.

The source index is defined using the explicit formula (2), with natural-number division. `Stern.h_source_exact` proves that its rational value is exactly (2); thus integer truncation does not alter the sequence. The recurrence leads to the concatenation and adjacent-value lemmas, then to `Stern.h_quadratic` and `Stern.h_sum_of_squares`. The principal theorem is

$$\texttt{Stern.h_positive}: \qquad \forall n \in \mathbb{N}\ \forall t \in \mathbb{R}, \quad 0 < B_{h_n}(t).$$

Its zero-parameter case and absence-of-real-roots corollary are also checked as `Stern.h_at_zero` and `Stern.h_no_real_root`.

The counterexample file checks the recurrence computation `Stern.B183_explicit`, the derivative formula, and `Stern.B183_deriv_neg_two_thirds`. It connects the negative derivative directly to `Stern.B183_not_monotoneOn` on $(-1, 0)$ by showing that a nondecreasing function's right-hand slopes cannot have a negative limit. Failure of strict increase and of global increase are also checked, as are the original-family specialization $s_{0,2} = 183$ and the exact two-point certificate `Stern.B183_`

`two_point_certificate`. Finally, `Stern.B183_zero_iff` proves that a real zero is exactly -1, using the verified factorization and positive quotient. The theorem `Stern.B183_deriv_neg_one` also checks $P'(-1) = 1$. The elementary simplicity argument in Proposition 7 is not separately encoded as a polynomial-root multiplicity theorem.

All five library modules, their aggregate module, and the axiom audit compiled successfully. No admitted proof or additional axiom is used. The printed axiom dependencies are subsets of the standard Lean dependencies `propext`, `Classical.choice`, and `Quot.sound`. Exact theorem statements, the tested commands, and their outputs are recorded in `STATUS.md`, `BUILD.log`, and `AXIOMS.log`. The local build used the installed toolchain and existing mathlib dependencies; a fresh network download of those dependencies was not tested. The supplementary Python checker uses exact arithmetic for finite checks, including $0 \le n \le 20$. Neither the natural-language proof nor the formal all-parameter theorem depends on those finite tests.

6 Sources, priority, and assistance

The conjecture statements and the quadratic identity were checked against the 17-page arXiv version [2]. The journal publication metadata are supplied in the bibliography, but access to its full text was restricted; page-by-page agreement with the journal version is not asserted. The accessible paper by Altizio [3] was also examined. Its stated results on zero-free complex regions and prime-index irreducibility do not by themselves establish the results proved here. The official record for Altizio's 2025 doctoral thesis [4] was available, but the thesis full text could not be accessed. Its coverage of the present questions is therefore undetermined. The literature search is not exhaustive, and this note does not assert that these conclusions are the first published resolutions. This work was prepared with AI assistance; no claim of human expert peer review is made.

References

[1] S. Klavžar, U. Milutinović, and C. Petr, *Stern polynomials*, Advances in Applied Mathematics **39** (2007), 86–95.

[2] M. Ulas, *Strong arithmetic property of certain Stern polynomials*, arXiv:1909.10844v1 (2019), `https://arxiv.org/abs/1909.10844v1`. Published in Publicationes Mathematicae Debrecen **96** (3–4) (2020), 401–422, doi:10.5486/PMD.2020.8683. Page references in this note are to the arXiv version.

[3] D. Altizio, *Zeros of Stern polynomials in the complex plane*, arXiv:2511.03847v1 (2025), `https://arxiv.org/abs/2511.03847v1`.

[4] D. Altizio, *On Stern polynomials and their zeroes*, Ph.D. thesis, University of Illinois Urbana-Champaign, 2025. Official record: `https://www.ideals.illinois.edu/items/136452`. Metadata consulted; full text unavailable for this note.